Spring 2020
Math 142
Exam 1
Form B

Last Name:_____ First Name:_____

UIN:_____ Section #:_____

On my honor, as an Aggie, I have neither given nor received unauthorized aid on this academic work.

Signature:_____

INSTRUCTIONS

- TURN OFF AND PUT AWAY ALL CELL PHONES AND ELECTRONIC DEVICES. IF AN ELECTRONIC DEVICE (OTHER THAN YOUR CALCULATOR) IS SEEN OR HEARD DURING THE EXAM YOU WILL RECEIVE A ZERO.
- The gray or blue Texas A&M University Scantron is required for this exam.
- Fill in your scantron with the appropriate information. To avoid a possible 10 point penalty, you must fill in and bubble your Last Name, First Name, MI, DEPT (MATH), Course No. (142), Section (505), UIN, and Test Form(A or B). Please also write "Arpan" in the Instructor field, "Exam 1" in the Exam Field, and then sign in the appropriate place.
- You will have 50 minutes to complete this exam. As soon as time is called, you must stop writing, put down your pencil, and close your exam.
- **Part 1: Multiple Choice (Problems 1-10)** Each problem is worth 6 points. You **MUST** record your answer on your scantron. It is highly recommended that you record your answer on your exam as well. Unless noted otherwise, answer choices are rounded to four decimal places.
- **Part 2: Workout (Problem 11-14)** You are required to show your work on these problems. An answer with no work is unacceptable. Please clearly box your final answer. Each workout problem is worth 10 points.

"An Aggie does not lie, cheat or steal or tolerate those who do."

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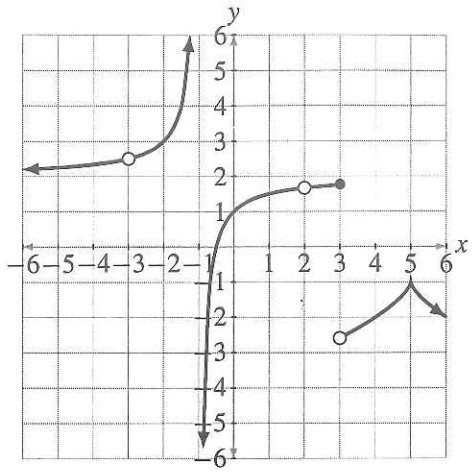
Part 1: Multiple-Choice Problems. Each multiple choice question is worth 6 points. You **MUST** record your answer on your scantron. It is highly recommended that you record your answer on your exam as well. Unless noted otherwise, answer choices are rounded to four decimal places.

1. Use the graph of $f(x)$ below to estimate $\lim_{x\to -3} f(x)$.



(a) $\lim_{x\to -3} f(x) = 2.$

(b) $\lim_{x\to -3} f(x)$ Does Not Exist.

(c) $\lim_{x\to -3} f(x) = -\infty.$

(d) $\lim_{x\to -3} f(x) = 1.75.$

(e) $\lim_{x\to -3} f(x) = 2.5.$

2. Find $\lim_{x\to 2} \frac{x+2}{x-2}$, if it exists. If it does not exist, use limits to describe the way in which it does not exist.

(a) The limit does not exist and $\lim_{x\to 2^-} f(x) \to \infty$ and $\lim_{x\to 2^+} f(x) \to -\infty$.

(b) The limit does not exist and $\lim_{x\to 2^-} f(x) \to -\infty$ and $\lim_{x\to 2^+} f(x) \to \infty$.

(c) The limit does not exist and $\lim_{x\to 2} f(x) \to \infty$

(d) The limit exists, and it equals 0.

(e) The limit exists, and it equals 1.

3. Complete the table below for $f(x) = \frac{x-3}{\ln(x-2)}$ and then use it to determine which of the following statements below is TRUE. (Answers below are rounded to six decimal places.)

x	2.9	2.99	3	3.01	3.1
$f(x)$					

(a) $f(3.1) = 1.000500$ and based on the table above it appears that $\lim_{x\to 3} f(x) \approx 1$.

(b) $f(3.01) = 1.004992$ and based on the table above it appears that $\lim_{x\to 1} f(x) \approx 3$.

(c) $f(2.99) = 1.233152$ and based on the table above it appears that $\lim_{x\to 1} f(x) \approx 3$.

(d) $f(2.9) = 0.949122$ and based on the table above it appears that $\lim_{x\to 3} f(x) \approx 1$.

(e) None of the above statements are true.

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4. Consider the function $f(x) = \begin{cases} x-2 & x \leq 2 \\ 2-x & x > 2 \end{cases}$. Mark the correct option.

   (a) $f(x)$ is differentiable at $x = 2$.

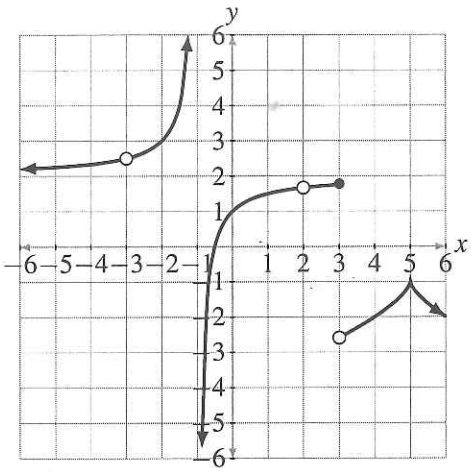
   (b) $f(x)$ is continuous but not differentiable at $x = 2$.

   (c) $f(x)$ is defined but not continuous at $x = 2$.

   (d) $f(x)$ is not defined at $x = 2$.

   (e) None of the above.

5. Here is the graph of $f(x)$. Which of the following statements below is TRUE?



   (a) The graph of $f(x)$ is continuous at $x = 5$.

   (b) The graph of $f(x)$ is continuous at $x = 3$.

   (c) The graph of $f(x)$ is continuous at $x = 2$.

   (d) The graph of $f(x)$ is continuous at $x = -1$.

   (e) The graph of $f(x)$ is continuous at $x = -3$.

6. What is the average rate of change of the function $f(x) = \dfrac{1}{x^2}$ on the interval $[1,2]$?

   (a) 0

   (b) -0.25

   (c) -0.5

   (d) -0.75

   (e) None of the above.

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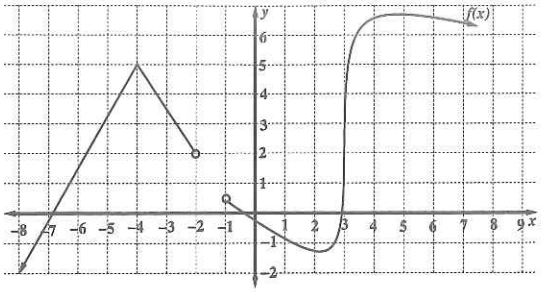
7. Given the graph of $f(x)$ below, for what value(s) of x is $f(x)$ differentiable?



(a) $(-\infty,-4)\cup(-4,-2)\cup(-1,3)\cup(3,\infty)$

(b) $(-\infty,-4)\cup(-4,-2)\cup(-2,\infty)$

(c) $(-\infty,-4)\cup(-4,-2)\cup(-1,\infty)$

(d) $(-\infty,-2)\cup(-1,\infty)$

(e) None of the above

8. Let $C(x)=x^{3/2}+12x^{1/2}+50$ be the cost in dollars for Sbisa Dining Hall to serve x people lunch in a day. Find the marginal cost function and use it to estimate the cost of the 37th lunch they make in a day.

(a) \$16

(b) \$14

(c) \$12

(d) \$10

(e) \$8

9. If $k(x)=3e^x-\sqrt{x^5}+\ln\left(\frac{2}{x^3}\right)-8^3$, find its derivative.

(a) $k'(x)=3e^x-\frac{5}{2}\sqrt{x^3}-\frac{3}{x}$

(b) $k'(x)=3e^x+\frac{5}{2}\sqrt{x^3}-3x$

(c) $k'(x)=3e^x-\frac{5}{2}\sqrt{x^3}-\frac{3}{x}-8^3$

(d) $k'(x)=3e^x+\frac{5}{2}\sqrt{x^3}-3x-8^3$

(e) None of the above.

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10. Find the equation of the tangent line to the curve $f(x) = 3x^2 - 7x + 4e^x$ at $x = 0$.

(a) $y = 3x + 11$

(b) $y = -3x + 7$

(c) $y = -3x + 4$

(d) $y = 3x + 7$

(e) None of the above

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STOP!! All of your answers up until this point need to be recorded on your scantron!

It is highly recommend that you record your answer on your exam as well but the answers marked on the scantron will take precedence if there is a discrepancy. Please note that the total number of points you received credit for on the scantron portion of the exam will be posted on eCampus but not written on this exam when it is returned to you.

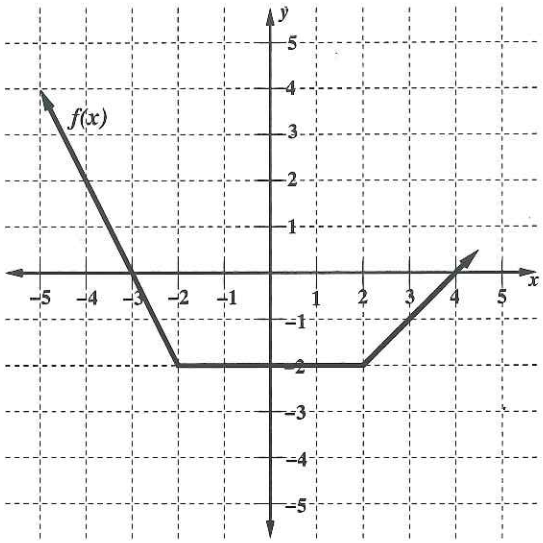
Part 2: Workout Problems. You are required to show your work on these problems. An answer with no work is unacceptable. Please clearly box your final answer. Each problem is worth 10 points.

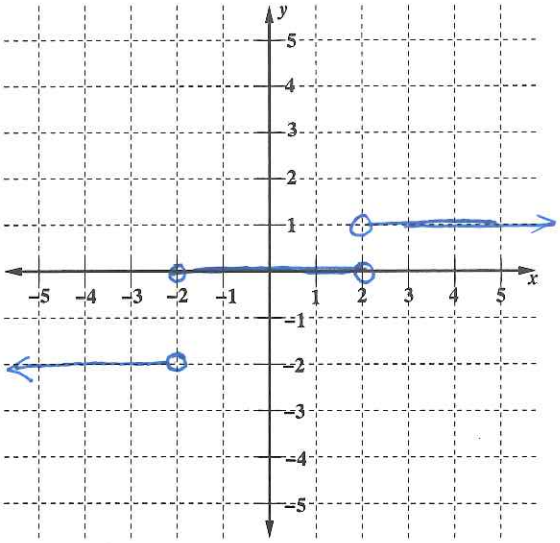
11. Consider the function $f(x) = \dfrac{x^2 e^x}{x^3 - 3}$. Find $f'(x)$ (Do not simplify).

$$f(x) = \frac{x^2 e^x}{x^3 - 3}$$

$$f'(x) = \frac{(x^3-3)(2x e^x + x^2 e^x) - x^2 e^x (3x^2)}{(x^3-3)^2}$$

12. Given the graph of $f(x)$ below, sketch a graph of $f'(x)$. Clearly label at least two points on the graph of $f'(x)$.





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13. Consider the function $f(x) = 4x^2 - 7x + 3$. Use the limit definition of the derivative to find $f'(x)$.

$$f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$$

$$= \lim_{h \to 0} \frac{4(x+h)^2 - 7(x+h) + 3 - (4x^2 - 7x + 3)}{h}$$

$$= \lim_{h \to 0} \frac{4(x^2 + 2xh + h^2) - 7x - 7h + 3 - (4x^2 - 7x + 3)}{h}$$

$$= \lim_{h \to 0} \frac{8xh + 4h^2 - 7h}{h} = \lim_{h \to 0} (8x + 4h - 7)$$

$$= \boxed{8x - 7}$$

14. Algebraically evaluate the limit below. You must show all of your work like we did in-class.

$$\lim_{x \to 5} \frac{x^2 - 2x - 15}{x^2 - 7x + 10}$$

$$= \lim_{x \to 5} \frac{x^2 - 5x + 3x - 15}{x^2 - 5x - 2x + 10}$$

$$= \lim_{x \to 5} \frac{x(x-5) + 3(x-5)}{x(x-5) - 2(x-5)}$$

$$= \lim_{x \to 5} \frac{(x-5)(x+3)}{(x-5)(x-2)} = \lim_{x \to 5} \frac{x+3}{x-2} = \boxed{\frac{8}{3}}$$

The total number of points you received credit for on the workout portion of the exam is _____ (out of a possible 40 points).

Please fill in these blank: Name:_____.